

Preface

Applied Analysis

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Chapter 1

Normed Linear Spaces

§ 1.1. Basic Definitions and Results

Definition 1.1.1. A seminorm on an $\mathbb{F}$ -vector space X is a map:

$$\|\cdot\| : X \rightarrow [0, \infty)$$

satisfying:

- (1) $\|\alpha x\| = |\alpha| \|x\|$ for all $\alpha \in \mathbb{F}$ and $x \in X$ (*homogeneity*);
- (2) $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in X$ (*triangle inequality*).

If the following condition is also satisfied:

- (3) $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = 0$ (*positive definiteness*),

then we say $\|\cdot\|$ is a norm. The ordered pair $(X, \|\cdot\|)$ is called a normed linear space.

Remark. In most cases, one can take $\mathbb{F}$ to be $\mathbb{R}$ or $\mathbb{C}$ without any loss of generality.

Remark. A normed linear space induces a metric by the equation $d(x, y) := \|y - x\|$.

Remark. A normed linear space is finite dimensional if it is as a vector space; otherwise it is infinite dimensional.

Definition 1.1.2. Let $(X, \|\cdot\|)$ be a normed linear space.

- (1) The open ball centered around $x_0 \in X$ with radius $r > 0$ is denoted $B(x_0, r) := \{x \in X \mid \|x - x_0\| < r\}$.
- (2) The closed ball centered around $x_0 \in X$ with radius $r > 0$ is denoted $\overline{B}(x_0, r) := \{x \in X \mid \|x - x_0\| \leq r\}$.
- (3) The sphere centered around $x_0 \in X$ with radius $r > 0$ is denoted $S(x_0, r) := \{x \in X \mid \|x - x_0\| = r\}$.

Definition 1.1.3. Two norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ on a normed linear space X are said to be equivalent if there exists a constant $C \geq 1$ such that, for all $x \in X$:

$$\frac{1}{C} \|x\|_\beta \leq \|x\|_\alpha \leq C \|x\|_\beta.$$

Example 1.1.1. Let $x \in \mathbb{R}^n$, where $x := (x_1, \dots, x_n)$. Then:

$$\begin{aligned}\|x\|_2 &:= \left(\sum_{i=1}^n |x_i|^2 \right)^{\frac{1}{2}}, \\ \|x\|_1 &:= \sum_{i=1}^n |x_i|, \\ \|x\|_\infty &:= \max_{i=1}^n |x_i|, \\ \|x\|_p &:= \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}, \quad p > 1\end{aligned}$$

are norms on $\mathbb{R}^n$.

Example 1.1.2. Let $a, b \in \mathbb{R}$ with $a < b$. The set of continuous real-valued functions on $[a, b]$ is denoted $C([a, b]) := \{f : [a, b] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$. Then:

$$\begin{aligned}\|f\|_\infty &:= \sup\{|f(x)| \mid x \in [a, b]\}, \\ \|f\|_2 &:= \left(\int_a^b |f(x)|^2 dx \right)^{\frac{1}{2}}, \\ \|f\|_p &:= \left(\int_a^b |f(x)|^p dx \right)^{\frac{1}{p}},\end{aligned}$$

are all norms on $C([a, b])$.

Definition 1.1.4. Let $(X, \|\cdot\|)$ be a normed linear space. A subset $V \subset X$ is convex if for all $x, y \in V$ and for all $t \in [0, 1]$, $tx + (1 - t)y \in V$.

Example 1.1.3. Any subspace of a normed linear space is convex. Moreover, by the triangle inequality any open ball is convex.

Remark. Note that the *supremum norm* $\|\cdot\|_\infty$ on $C([a, b])$ only exists (by the Extreme Value Theorem) because $[a, b]$ is compact. Hence the above example can be generalized to a compact metric space Y . In particular, for $C(Y) := \{f : Y \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$, the norm $\|f\|_\infty := \sup\{|f(x)| \mid x \in Y\}$ always exists and is finite.

Proposition 1.1.1. Let (X, d) and (Y, ρ) be metric spaces. Let $f : X \rightarrow Y$. Then f is continuous if and only if the preimage of an open set in Y is open in X .

Proof. Suppose f is continuous. Let $V \subseteq Y$ be open. If $f^{-1}(V) = \emptyset$, then there is nothing to show. If not, pick any $x \in f^{-1}(V)$. Then $f(x) \in V$. Since V is open, there exists $\epsilon > 0$

such that $f(x) \in B(f(x), \delta) \subseteq V$. Since f is continuous, there exists a $\delta > 0$ such that $f(B(x, \delta)) \subseteq B(f(x), \epsilon) \subseteq V$. Thus $B(x, \delta) \subseteq f^{-1}(V)$; i.e., $f^{-1}(V)$ is open.

Conversely, let $x \in X$ and $\epsilon > 0$. Since $B(f(x), \epsilon)$ is open in Y , $f^{-1}(B(f(x), \epsilon))$ is open in X . Clearly $x \in f^{-1}(B(f(x), \epsilon))$. Since this set is open, there exists $\delta > 0$ such that $x \in B(x, \delta) \subseteq f^{-1}(B(f(x), \epsilon))$. Hence $f(B(x, \delta)) \subseteq B(f(x), \epsilon)$. Since the set of all open balls is a basis for a topology, any open set can be written as the union of open balls. Thus f is continuous. $\square$

Remark. It follows that f is continuous if and only if the preimage of a closed set is closed.

Definition 1.1.5. Let $(X, \|\cdot\|)$ be a normed linear space and $\{x_n\}_{n=1}^{\infty}$ be a sequence in X .

- (1) The sequence $\{x_n\}_{n=1}^{\infty}$ converges to $x \in X$ in the $\|\cdot\|$ -norm if $\|x - x_n\| \rightarrow 0$.
- (2) A series in X is a formal sum $\sum_{n=1}^{\infty} x_n$. We say the series $\sum_{n=1}^{\infty} x_n$ converges to $x \in X$ if the sequence of partial sums $\{\sum_{i=1}^n x_i\}_{n=1}^{\infty}$ converges to x . We say $\sum_{n=1}^{\infty} x_n$ is absolutely convergent if $\sum_{n=1}^{\infty} \|x_n\|$ converges.

Proposition 1.1.2. Let X be a normed linear space. The following are equivalent:

- (1) X is a Banach space;
- (2) Every absolutely convergent series in X converges.

Proof.

$\square$

§ 1.2. Completeness and Compactness

Lemma 1.2.1. In $(\mathbb{R}^n, \|\cdot\|_{\infty})$, every closed ball $\overline{B}(0, c)$ is compact.

Proof.

$\square$

Theorem 1.2.1. Let X be a finite dimensional normed linear space. Every closed and bounded subset of X is compact.

Proof.

$\square$

Corollary 1.2.1. Every finite dimensional normed linear space is Banach.

Proof. Let X be a finite dimensional normed linear space and $\{x_n\}_{n=1}^{\infty}$ a Cauchy sequence. Then $\{x_n\}_{n=1}^{\infty}$ is closed and bounded, hence it is compact. Thus $\{x_n\}_{n=1}^{\infty}$ admits a convergent subsequence. But since $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence which admits a convergent subsequence, $\{x_n\}_{n=1}^{\infty}$ is convergent. Thus X is Banach. $\square$

Corollary 1.2.2. *Every finite dimensional subspace of a normed linear space is closed.*

Proof. This follows from Corollary 1.2.1 and Exercise 1.2.4. $\square$

Lemma 1.2.2 (Riesz's Lemma). *Let $(X, \|\cdot\|)$ be a normed linear space. Let Z be a closed and proper subspace of X . Let $0 < \lambda < 1$ be fixed. Then there exists some $x \in X \setminus Z$ such that $\|x\| = 1$ and $\text{dist}(x, Z) > \lambda$.*

Proof. $\square$

Theorem 1.2.2. *Let X be a normed linear space. Then X is finite dimensional if and only if $\overline{B}(0, 1)$ is compact.*

Proof. $\square$

Exercises

Exercise 1.2.1. *Let K be a compact metric space (fine to assume that $K = [0, 1]$). Show that $C(K)$ paired with the sup-norm $\|\cdot\|_{\infty}$ is Banach.*

Proof. $\square$

Exercise 1.2.2. *In any metric space, a closed subset of a compact set is compact.*

Proof. $\square$

Exercise 1.2.3. *In any metric space, a Cauchy sequence which has a convergent subsequence must converge.*

Proof.

□

Exercise 1.2.4. *A subspace of a Banach space is closed if and only if it is Banach.*

Proof. Let X be a Banach space and Y a subspace. Suppose that Y is closed. Let $\{y_n\}_{n=1}^{\infty} \subseteq Y$ be a Cauchy sequence. Then $\{y_n\}_{n=1}^{\infty}$ is Cauchy in X . Therefore $y_n \rightarrow x$ for some $x \in X$. But since Y is closed, it contains all of its limit points. Hence $x \in Y$. Therefore $\{y_n\}_{n=1}^{\infty}$ converges in Y , establishing that Y is Banach.

Conversely, suppose that Y is Banach. Let $\{y_n\}_{n=1}^{\infty} \subseteq Y$ be a sequence such that $y_n \rightarrow x \in X$. Note that $\{y_n\}_{n=1}^{\infty}$ is convergent, hence Cauchy. But since Y is Banach, $\{y_n\}_{n=1}^{\infty}$ converges to some $y \in Y$. It must be the case that $x = y \in Y$. Therefore Y contains all of its limit points; i.e., it is closed in X . □

Exercise 1.2.5. *In any normed linear space X with closed subspace Z , if $x \in X \setminus Z$ then $\text{dist}(x, Z) > 0$.*

Proof. Suppose towards contradiction $\text{dist}(x, Z) = \inf\{\|x - z\| \mid z \in Z\} = 0$. We can find a sequence $\{z_n\}_{n=1}^{\infty} \subseteq Z$ such that $\|x - z_n\| \rightarrow 0$. Hence $z_n \rightarrow x$. But since Z is closed, $x \in Z$. This is a contradiction, hence $\text{dist}(x, Z) > 0$. □

Exercise 1.2.6. *In any normed linear space X with subspace Z , if $x \in X$ and $z \in Z$ then $\text{dist}(x, Z) = \text{dist}(x + z, Z)$.*

Proof. Note that $\text{dist}(x + z, Z) = \inf\{\|(x + z) - z_0\| \mid z_0 \in Z\} = \inf\{\|x - (z_0 - z)\| \mid z_0 \in Z\}$. Since $\{z_0 - z \mid z_0 \in Z\} = Z$, we have that $\inf\{\|x - z\| \mid z \in Z\} = \inf\{\|(x + z) - z_0\| \mid z_0 \in Z\}$; i.e., $\text{dist}(x, Z) = \text{dist}(x + z, Z)$. □

Exercise 1.2.7. *Let X and Y be normed linear spaces and let $\varphi : X \rightarrow Y$ be a linear operator. Show that φ is continuous if and only if it is continuous at 0.*

Proof.

□

§ 1.3. Linear Operators

Definition 1.3.1. Let X and Y be normed linear spaces. A map $T : X \rightarrow Y$ is linear if $T(x + \alpha y) = T(x) + \alpha T(y)$ for all $x, y \in X$ and $\alpha \in \mathbb{F}$. T is called a linear transformation (or linear operator). If X is a normed linear space over $\mathbb{F}$, then a linear transformation $\varphi : X \rightarrow \mathbb{F}$ is called a linear functional.

Remark. The norm on $\mathbb{F}$ is the usual absolute value in $\mathbb{R}$ or modulus operator in $\mathbb{C}$.

Example 1.3.1. Linear operators between $\mathbb{R}^n$ and $\mathbb{R}^m$ correspond to $n \times m$ matrices.

Example 1.3.2. Consider the normed linear space $(C([0, 1]), \|\cdot\|_\infty)$. Evaluation at zero is a linear functional, as well as the map $x \mapsto \int_0^1 x(t)dt$. Moreover, $T : C([0, 1]) \rightarrow C([0, 1])$ defined by $x(t) \mapsto \int_0^1 K(s, t)x(s)ds$ is a linear transformation, where K is a continuous function on $[0, 1] \times [0, 1]$.

Definition 1.3.2. Let X and Y be normed linear spaces and $T : X \rightarrow Y$ a linear transformation. T is bounded if there exists $M > 0$ such that for all $x \in X$, $\|x\| \leq 1$ implies $\|Tx\| \leq M$.

Theorem 1.3.1. Let X and Y be normed linear spaces. A linear transformation $T : X \rightarrow Y$ is bounded if and only if it is continuous.

Proof. Suppose T is continuous. Then T is continuous at zero. Choose $\delta > 0$ such that $\|x\| < \delta$ implies $\|Tx\| < 1$. Now let $x \in X$ with $\|x\| \leq 1$. Then $\|Tx\| = \left\|T\left(\frac{x\delta}{\delta}\right)\right\| = \left|\frac{1}{\delta}\right| \|T(\delta x)\| \leq \frac{1}{\delta}$. Therefore T is bounded.

Conversely, let $M > 0$ satisfy $\|Tx\| \leq M$ for all $x \in \bar{B}(0, 1)$. Let $\epsilon > 0$. Choose $\delta = \frac{\epsilon}{M}$. Then $\|x\| \leq \delta$ implies $\left\|\frac{M}{\epsilon}x\right\| \leq 1$. Hence by our assumption $\left\|T\left(\frac{M}{\epsilon}x\right)\right\| \leq M$, which is equivalent to $\|Tx\| \leq \epsilon$. Thus T is continuous at 0, hence it is continuous everywhere. $\square$

Example 1.3.3. Consider $D : C^1([0, 1]) \rightarrow C([0, 1])$ given by $Df = f'$. Consider the sequence of functions $\{f_n\}_{n=1}^\infty \subseteq C^1([0, 1])$ given by $f_n(x) = \frac{1}{\sqrt{n}}x^n$. Then for all $n \in \mathbb{N}$, $Df_n(x) = \sqrt{n}x^{n-1}$. Hence D is unbounded.

Theorem 1.3.2. Let X be a normed linear space. A linear functional $\varphi : X \rightarrow \mathbb{F}$ is continuous if and only if its kernel is closed.

Proof. Suppose φ is continuous. Then $\ker \varphi = \varphi^{-1}(\{0\})$ is closed, as the continuous preimage of a closed set is closed.

Conversely, suppose towards contradiction φ is not continuous. Then φ is unbounded; i.e., there exists a sequence $\{x_n\}_{n=1}^\infty \subset X$ such that $\|x_n\| = 1$ for all $n \in \mathbb{N}$ and $|\varphi(x_n)| \rightarrow \infty$. If $\ker \varphi = X$ then there is nothing to show. Assume $\ker \varphi \neq X$ and pick some $x \in X \setminus \ker \varphi$. Define $y_n := x - \frac{\varphi(x)}{\varphi(x_n)}x_n$. Then $\varphi(y_n) = \varphi(x) - \frac{\varphi(x)}{\varphi(x_n)}\varphi(x_n) = 0$. Hence $y_n \in K$ for all $n \in \mathbb{N}$. But note that $\|y_n - x\| = \left\|-\frac{\varphi(x)}{\varphi(x_n)}x_n\right\| = \frac{|\varphi(x)|}{|\varphi(x_n)|}\|x_n\| \rightarrow 0$. Hence $y_n \rightarrow x$. Since $\ker \varphi$ is closed, $x \in \ker \varphi$. This is a contradiction, establishing the proof. $\square$

Theorem 1.3.3. Let X be a finite dimensional normed linear space and Y a normed linear space. If $T : X \rightarrow Y$ is linear, then T is continuous.

Proof. Pick a basis for X such that $X = \text{span}\{b_1, \dots, b_n\}$. Define $M := \max_{i=1}^n \{T(b_i)\}$. For each $j = 1, \dots, n$, define a linear functional $\varphi_j : X \rightarrow \mathbb{F}$ by:

$$\begin{aligned}\varphi_j(x) &= \varphi_j\left(\sum_{i=1}^n \alpha_i b_i\right) \\ &= \sum_{i=1}^n \alpha_i \varphi_j(b_i) \\ &:= \alpha_j\end{aligned}$$

Since all subspaces of X are closed by Corollary 1.2.2, surely $\ker \varphi_j$ is closed. Hence by Theorem 1.3.2 φ_j is continuous. Since φ_j is continuous, find $L > 0$ sufficiently large enough such that for every $x \in X$ with $\|x\| \leq 1$, we have $|\varphi_j(x)| \leq L$. Since $T : X \rightarrow Y$ is linear:

$$\begin{aligned}\|Tx\| &= \left\| T \sum_{i=1}^n \alpha_i b_i \right\| \\ &= \left\| \sum_{i=1}^n \alpha_i T b_i \right\| \\ &= \left\| \sum_{i=1}^n \varphi_i(x) T b_i \right\| \\ &\leq \sum_{i=1}^n |\varphi_i(x)| \|T b_i\| \\ &\leq nLM \\ &< \infty.\end{aligned}$$

Thus T is continuous. □

Theorem 1.3.4. *Let X be a finite dimensional vector space. All norms on X are equivalent.*

Proof. Let $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ be two norms. Consider $\text{id} : (X, \|\cdot\|_\alpha) \rightarrow (X, \|\cdot\|_\beta)$. This map is surely linear, hence by Theorem 1.3.3 id is continuous. Hence we can find some $A > 0$ sufficiently large such that, for all $x \in X$ with $\|x\|_\alpha \leq 1$, we have $\|x\|_\beta \leq A$. Thus:

$$\begin{aligned}\|x\|_\beta &= \left\| \|x\|_\alpha \frac{x}{\|x\|_\alpha} \right\|_\beta \\ &= \|x\|_\alpha \left\| \frac{x}{\|x\|_\alpha} \right\|_\beta \\ &\leq \|x\|_\alpha A.\end{aligned}$$

Applying the same argument to id^{-1} gives $\|x\|_\alpha \leq \|x\|_\beta B$, establishing the theorem. □

Exercises

Exercise 1.3.1. Let X be a normed linear space. Prove the limit laws: if $x_n \rightarrow x$, $y_n \rightarrow y$, and $\lambda \in \mathbb{F}$, then $x_n + \lambda y_n \rightarrow x + \lambda y$.

Proof.

□

Exercise 1.3.2. Let X be a normed linear space with norm $\|\cdot\|$. Prove that the map $\|\cdot\| : X \rightarrow \mathbb{R}$ is continuous.

Proof. Let $x, y \in X$. Then $\|x\| \leq \|x - y\| + \|y\|$, giving $\|x\| - \|y\| \leq \|x - y\|$. A similar tactic gives $\|y\| - \|x\| \leq \|x - y\|$. Hence $|\|x\| - \|y\|| \leq \|x - y\|$. This means $\|\cdot\|$ is Lipschitz, implying that it is continuous. □

Exercise 1.3.3. Prove that the closure of a linear subspace in a normed linear space is also a subspace.

Proof. Let X be a normed linear space and Y a subspace. Let $u, v \in \bar{Y}$ and $\alpha \in \mathbb{F}$. There exists sequences $\{u_n\}_{n=1}^\infty, \{v_n\}_{n=1}^\infty$ such that $u_n \rightarrow u$ and $v_n \rightarrow v$. Hence $u_n + \alpha v_n \rightarrow u + \alpha v \in \bar{Y}$. Thus $\bar{Y}$ is a linear subspace. □

Exercise 1.3.4. Prove that the operator norm has the three properties required of a norm

Proof.

□

Exercise 1.3.5. Prove directly that if T is an unbounded linear operator, then it is discontinuous at 0.

Proof. Let $T : X \rightarrow Y$ be an unbounded linear operator. Since T is unbounded:

$$\|T\|_{\mathcal{L}(X,Y)} = \sup\{\|Tx\|_Y \mid x \in \bar{B}_X(0,1)\} = \infty.$$

We can find a sequence $\{x_n\}_{n=1}^\infty$ such that $\|x_n\|_X \leq 1$ and $\|Tx_n\|_Y \rightarrow \infty$. Consider the sequence $\left\{\frac{x_n}{\|Tx_n\|_Y}\right\}_{n=1}^\infty$. Clearly $\frac{x_n}{\|Tx_n\|_Y} \rightarrow 0$. But:

$$\left\|T\left(\frac{x_n}{\|Tx_n\|_Y}\right)\right\| = \frac{\|Tx_n\|_Y}{\|Tx_n\|_Y} = 1.$$

Hence the sequence $\left\{T\left(\frac{x_n}{\|Tx_n\|_Y}\right)\right\}_{n=1}^{\infty}$ does not converge to 0. Thus T is discontinuous at 0. $\square$

Exercise 1.3.6. Prove that the norm of a linear transformation is the infimum of all numbers M that satisfy the inequality $\|Tx\| \leq M\|x\|$ for all x .

Proof. Let M be such that $\|Tx\| \leq M\|x\|$ for all $x \in X$. Then $\frac{\|Tx\|}{\|x\|} \leq M$ for all $x \in X$, $x \neq 0$. Taking the supremum over all $x \in X$, $x \neq 0$ gives $\|T\|_{\mathcal{L}(X,Y)} \leq M$. Hence $\|T\|_{\mathcal{L}(X,Y)} \leq \inf\{M \mid \|Tx\| \leq M\|x\| \text{ for all } x \in X\}$.

Conversely, let $x \in X$, $x \neq 0$. Then $\|Tx\| \leq \|T\|_{\mathcal{L}(X,Y)}\|x\|$. Thus $\|T\|_{\mathcal{L}(X,Y)} \in \{M \mid \|Tx\| \leq M\|x\| \text{ for all } x \in X\}$. Surely $\|T\|_{\mathcal{L}(X,Y)} \geq \inf\{M \mid \|Tx\| \leq M\|x\| \text{ for all } x \in X\}$, establishing equality. $\square$

Exercise 1.3.7. On the space $C([0,1])$ we define "point-evaluation functionals" by $t^*(f) = f(t)$. Here $t \in [0,1]$ and $f \in C([0,1])$. Prove that $\|t^*\| = 1$. Prove that if $\varphi = \sum_{i=1}^n \lambda_i t_i^*$, where $t_1, t_2, \dots, t_n$ are distinct points in $[0,1]$, then $\|\varphi\| = \sum_{i=1}^n |\lambda_i|$.

Proof. $\square$

Exercise 1.3.8. Prove that if a linear transformation maps some nonvoid open set of the domain space to a bounded set in the range space, then it is continuous.

Proof. Let $T : X \rightarrow Y$ be linear. Let $U \subseteq X$ be open and nonvoid. Then there exists some $x \in U$. Find $r > 0$ such that $B(x, r) \subseteq U$.

Let $x_0 \in B(0, 1)$. Then $x + rx_0 \in B(x, r) \subseteq U$. Hence $T(x + rx_0) \in T(U)$. We can write:

$$\|Tx_0\| \leq \frac{\|T(x + rx_0)\| + \|Tx\|}{r}.$$

Since $T(U)$ is bounded, it must be the case that $\|Tx_0\| \leq M$ for some $M > 0$. Thus $T(B(0, 1))$ is bounded.

Now let $x \in X$. We have that:

$$\begin{aligned} \|Tx\| &= \left\| T\left(\frac{x}{2\|x\|} \cdot 2\|x\|\right) \right\| \\ &= 2 \left\| T\left(\frac{x}{2\|x\|}\right) \right\| \|x\| \\ &\leq 2M \|x\|. \end{aligned}$$

Thus T is continuous. $\square$

§ 1.4. Bounded Linear Functionals

Definition 1.4.1. Let X and Y be normed linear spaces.

- (1) The space of bounded linear operators between X and Y is denoted $\mathcal{L}(X, Y) := \{T : X \rightarrow Y \mid T \text{ is bounded}\}$.
- (2) The operator norm of a bounded linear operator $T \in \mathcal{L}(X, Y)$ is $\|T\|_{\mathcal{L}(X, Y)} := \sup\{\|Tx\|_Y \mid x \in \overline{B}_X(0, 1)\}$.
- (3) The continuous dual of X is defined as $X^* := \mathcal{L}(X, \mathbb{F})$.

Remark. We denote $\mathcal{L}(X, X)$ by $\mathcal{L}(X)$.

Proposition 1.4.1. If $T \in \mathcal{L}(X, Y)$ and $x \in X$, then $\|Tx\| \leq \|T\|_{\mathcal{L}(X, Y)} \cdot \|x\|$.

Proof. If $x = 0$, then there is nothing to show. If $x \neq 0$, then:

$$\begin{aligned} \|Tx\| &= \left\| \|x\| T\left(\frac{x}{\|x\|}\right) \right\| \\ &= \left\| T\left(\frac{x}{\|x\|}\right) \right\| \|x\| \\ &\leq \|T\|_{\mathcal{L}(X, Y)} \|x\|. \end{aligned} \quad \square$$

Proposition 1.4.2. If $S, T \in \mathcal{L}(X)$, then $S \circ T \in \mathcal{L}(X)$.

Proof. Let $x \in X$ such that $\|x\| \leq 1$. By Proposition 1.4.1:

$$\begin{aligned} \|(S \circ T)(x)\| &\leq \|S(T(x))\| \\ &\leq \|S\|_{\mathcal{L}(X)} \|Tx\| \\ &\leq \|S\|_{\mathcal{L}(X)} \|T\|_{\mathcal{L}(X)} \|x\| \\ &\leq \|S\|_{\mathcal{L}(X)} \|T\|_{\mathcal{L}(X)}. \end{aligned}$$

Taking the supremum over all $\|x\| \leq 1$ gives $\|S \circ T\|_{\mathcal{L}(X)} \leq \|S\|_{\mathcal{L}(X)} \|T\|_{\mathcal{L}(X)}$. $\square$

Theorem 1.4.1. Let X be a normed linear space and Y a Banach space. Then $(\mathcal{L}(X, Y), \|\cdot\|_{\mathcal{L}(X, Y)})$ is Banach.

Proof. Let $\{T_n\}_{n=1}^\infty \subseteq \mathcal{L}(X, Y)$ be Cauchy. Note that for any $x \in X$, $\{T_n(x)\}_{n=1}^\infty$ will be Cauchy in Y . Since Y is complete, this sequence has a limit. Define $Tx := \lim_{n \rightarrow \infty} T_n x$. We must show that T is linear, bounded, and $T_n \rightarrow T$.

Let $x, y \in X$, $\alpha \in \mathbb{F}$, and $n \in \mathbb{N}$. Then $T_n(x + \alpha y) = T_n x + \alpha T_n y$. By limit laws, $T(x + \alpha y) = Tx + \alpha Ty$. Thus T is linear.

Since $\{T_n\}_{n=1}^\infty$ is Cauchy in $\mathcal{L}(X, Y)$, this means $\{T_n\}_{n=1}^\infty$ is bounded. Hence we can find $M > 0$ sufficiently large such that $\|T_n\| \leq M$ for all $n \in \mathbb{N}$. For every $x \in X$ with $\|x\| \leq 1$, we have:

$$\begin{aligned} \|Tx\| &= \left\| \lim_{n \rightarrow \infty} T_n x \right\| \\ &= \lim_{n \rightarrow \infty} \|T_n x\| \\ &\leq \lim_{n \rightarrow \infty} \|T_n\|_{\mathcal{L}(X, Y)} \|x\| \\ &\leq \lim_{n \rightarrow \infty} \|T_n\|_{\mathcal{L}(X, Y)} \\ &\leq M. \end{aligned}$$

Thus T is bounded.

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ large such that $n, m \geq N$ implies $\|T_n - T_m\|_{\mathcal{L}(X, Y)} < \epsilon$. Let $x \in X$ such that $\|x\| \leq 1$. Then:

$$\begin{aligned} \|T_n x - T_m x\| &\leq \|(T_n - T_m)x\| \\ &\leq \|T_n - T_m\|_{\mathcal{L}(X, Y)} \|x\| \\ &\leq \|T_n - T_m\|_{\mathcal{L}(X, Y)} \\ &< \epsilon. \end{aligned}$$

Take $m \rightarrow \infty$. Then $\|T_n x - Tx\| \leq \epsilon$. Hence $\|(T_n - T)x\| \leq \epsilon$. Taking the supremum over all $x \in \overline{B}_X(0, 1)$ gives $\|T_n - T\|_{\mathcal{L}(X, Y)} \leq \epsilon$. Thus $T_n \rightarrow T$ in the operator norm. $\square$

Question. For any normed linear space X , is X^* nonempty? How can one construct an arbitrary bounded linear functional on X ?

Lemma 1.4.1 (Zorn's Lemma). *In a partially ordered set, if every chain has a maximal element, then the partially ordered set has a maximal element.*

Theorem 1.4.2 (Hahn-Banach). *Let X be a $\mathbb{R}$ -vector space and $p : X \rightarrow \mathbb{R}$ a map satisfying:*

$$\begin{aligned} p(x + y) &\leq p(x) + p(y), \\ p(\lambda x) &= \lambda p(x), \text{ provided } \lambda > 0. \end{aligned}$$

Let $Y \subseteq X$ be a subspace and $\varphi : Y \rightarrow \mathbb{R}$ a linear functional with:

$$\varphi(y) \leq p(y) \text{ for all } y \in Y.$$

Then there exists a linear functional $\Phi : X \rightarrow \mathbb{R}$ such that $\Phi|_Y = \varphi$ and:

$$\Phi(x) \leq p(x) \text{ for all } x \in X.$$

Proof.

□

Corollary 1.4.1. *Let X be a normed linear space and $Y \subseteq X$ a subspace. If $\varphi \in Y^*$, then there exists an extension $\Phi \in X^*$ such that $\|\varphi\|_{Y^*} = \|\Phi\|_{X^*}$.*

Proof. Define $p : X \rightarrow \mathbb{R}$ by $p(x) := \|\varphi\|_{Y^*} \|x\|$. Clearly $p(x + y) \leq p(x) + p(y)$ and $p(\lambda x) = \lambda p(x)$ provided $\lambda > 0$. Moreover:

$$\begin{aligned} p(y) &= \|\varphi\|_{Y^*} \|y\| \\ &\geq |\varphi(y)| \\ &\geq \varphi(y). \end{aligned}$$

By the Hahn-Banach Theorem, there exists an extension $\Phi : X \rightarrow \mathbb{R}$ such that $\Phi|_Y = \varphi$ and $\Phi(x) \leq p(x)$. From this, we have the following inclusion of sets:

$$\left\{ \left| \varphi\left(\frac{y}{\|y\|}\right) \right| : y \in Y, y \neq 0 \right\} \subseteq \left\{ \left| \Phi\left(\frac{x}{\|x\|}\right) \right| : x \in X, x \neq 0 \right\}.$$

Hence $\|\varphi\|_{Y^*} \leq \|\Phi\|_{X^*}$. Conversely, by our construction of p we have:

$$\begin{aligned} |\Phi(x)| &\leq |p(x)| \\ &\leq \|\varphi\|_{Y^*} \|x\|. \end{aligned}$$

Rearranging gives $\left| \Phi\left(\frac{x}{\|x\|}\right) \right| \leq \|\varphi\|_{Y^*}$. Taking the supremum over all $x \in X, x \neq 0$ gives $\|\Phi\|_{X^*} \leq \|\varphi\|_{Y^*}$. Having established equality, we've finally shown $\Phi \in X^*$. □

Corollary 1.4.2. *Let X be a normed linear space and $Y \subseteq X$ a subspace. If $w \in X \setminus Y$ such that $\text{dist}(w, Y) > 0$, then there exists a bounded linear functional $\varphi \in X^*$ such that $\varphi(w) = 1$, $\varphi(y) = 0$ for all $y \in Y$, and $\|\varphi\|_{X^*} = \frac{1}{\text{dist}(w, Y)}$.*

Proof. Define $\varphi : \text{span}\{Y, \{w\}\} \rightarrow \mathbb{R}$ by $\varphi(y + \lambda w) = \lambda$. This is a linear functional, and:

$$\begin{aligned} \|\varphi\| &= \sup \left\{ \left| \varphi\left(\frac{y + \lambda w}{\|y + \lambda w\|}\right) \right| : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0 \right\} \\ &= \sup \left\{ \frac{|\lambda|}{\|y + \lambda w\|} : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0 \right\} \\ &= \sup \left\{ \frac{1}{\left\| \frac{y}{\lambda} + w \right\|} : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0 \right\} \\ &= \sup \left\{ \frac{1}{\|w - y\|} : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0 \right\}^1 \\ &= \frac{1}{\inf\{\|w - y\| \mid y \in Y\}}^2 \\ &= \frac{1}{\text{dist}(w, Y)}. \end{aligned}$$

Applying Corollary 1.4.1 establishes the proof. $\square$

Remark. The previous corollary establishes that if $X = \{0\}$, then $X^* \neq \{0\}$.

Exercises

Exercise 1.4.1. Let Y be a linear subspace in a normed linear space X . Prove that:

$$\text{dist}(x, Y) = \sup\{\varphi(x) \mid \varphi \in X^*, \varphi|_Y = 0, \|\varphi\| = 1\}.$$

§ 1.5. Sequence Spaces

Exercises

Exercise 1.5.1. Show that $(\ell^\infty, \|\cdot\|_\infty)$ is complete.

Proof. $\square$

Exercise 1.5.2. Show that c and c_0 are closed in ℓ^∞ . Show that c_{00} is not closed by exhibiting a sequence in c_{00} which converges to a point $x \in \ell^\infty$ but is not in c_{00} .

Proof. $\square$

Exercise 1.5.3. Let $y \in \ell^\infty$ and define the linear functional:

$$\varphi_y : \ell^1 \rightarrow \mathbb{F}, \quad \varphi_y(x) := \sum_{j=1}^{\infty} x_j y_j.$$

Show that $\|\varphi_y\|_{(\ell^1)^*} = \|y\|_{\ell^\infty}$.

Proof. $\square$

¹Since $\{\frac{y}{\lambda} \mid y \in Y, \lambda \in \mathbb{R}\} = Y$.

²Let $x_n \rightarrow \inf A$. Then $\frac{1}{x_n} \rightarrow \frac{1}{\inf A}$. Since $\frac{1}{\inf A}$ is an upper-bound and an adherent point of $\frac{1}{A}$, we have $\frac{1}{\inf A} = \sup \frac{1}{A}$.

§ 1.6. Consequences of The Baire Category Theorem

Theorem 1.6.1 (Baire Category Theorem). *Let X be a complete metric space. If $\{U_n\}_{n=1}^{\infty}$ is a family of open and dense subsets of X , then $\bigcap_{n=1}^{\infty} U_n$ is still dense.*

Proof.

□

Corollary 1.6.1. *Let X be a complete metric space and $\{Z_n\}_{n=1}^{\infty}$ a family of closed sets. If $X = \bigcup_{n=1}^{\infty} Z_n$, then there exists some $i \in \mathbb{N}$ such that Z_i has non-empty interior.*

Proof.

□

Definition 1.6.1. Let X be a metric space.

- (1) A subset Y of X is called nowhere dense if $\bar{Y}$ has empty interior.
- (2) A subset of X is called Category I if it is the countable union of nowhere dense sets.
- (3) A subset of X which is not Category I is called Category II.

Remark. Theorem 1.6.1 and Corollary 1.6.1 can be rephrased as the assertion that every complete metric space is Category II.

Theorem 1.6.2 (Banach-Steinhouse). *Let X be a Banach space and Y a normed linear space. Let $\{T_{\alpha}\}_{\alpha \in A} \subset \mathcal{L}(X, Y)$ be an arbitrary family of bounded linear operators. Then $\sup\{\|T_{\alpha}\| \mid \alpha \in A\} < \infty$ if and only if $\{x \mid \sup_{\alpha \in A} \|T_{\alpha}x\| < \infty\}$ is Category II*

Proof. Suppose $C := \sup\{\|T_{\alpha}\| \mid \alpha \in A\} < \infty$. Then for all $x \in X$ and $\alpha \in A$, we have $\|T_{\alpha}x\|_Y \leq \|T_{\alpha}\| \|x\|_X \leq C \|x\|$. Hence $\{x \mid \sup_{\alpha \in A} \|T_{\alpha}x\| < \infty\} = X$, which is Category II by the Baire Category Theorem.

Conversely, suppose $F := \{x \mid \sup_{\alpha \in A} \|T_{\alpha}x\| < \infty\}$ is Category II. Define $F_n := \{x \mid \sup\{\|T_{\alpha}x\| \mid \alpha \in A\} \leq n\}$. This set is closed: **i cant figure out why, something about limits and supremum**. Since $F = \bigcup_{n=1}^{\infty} F_n$, by Corollary 1.6.1 there exists some $n \in \mathbb{N}$ such that F_n has nonempty interior. Choose $x_0 \in X$ and $r > 0$ such that $\bar{B}(x_0, r) \subset F_n$. Let $x \in X$ such

that $\|x\| \leq 1$. Fix $\alpha \in A$. Note that $x_0 + rx \in \overline{B}(x_0, r)$, hence:

$$\begin{aligned} \|T_\alpha x\|_Y &= \left\| T_\alpha \left(\frac{1}{r} (x_0 + rx - x_0) \right) \right\|_Y \\ &\leq \frac{1}{r} (\|T_\alpha(x_0 + rx)\|_Y + \|T_\alpha x_0\|_Y) \\ &\leq \frac{2n}{r} \\ &< \infty. \end{aligned}$$

Taking the supremum over all α and $\|x\| \leq 1$ gives $\sup\{\|T_\alpha\| \mid \alpha \in A\} < \infty$. $\square$

Corollary 1.6.2 (Uniform Boundedness Principle). *Let X be a Banach space and Y a normed linear space. Let $\{T_\alpha\}_{\alpha \in A} \subset \mathcal{L}(X, Y)$ be an arbitrary family of bounded linear operators. If $\sup_{\alpha \in A} \|T_\alpha x\|_Y < \infty$ for all $x \in X$, then $\sup_{\alpha \in A} \|T_\alpha\| < \infty$.*

Proof. The set $\{x \mid \sup_{\alpha \in A} \|T_\alpha x\| < \infty\} = X$ is Category II. By Theorem 1.6.2 we have that $\sup_{\alpha \in A} \|T_\alpha\| < \infty$. $\square$

Definition 1.6.2. Let X and Y be normed linear spaces and $T : X \rightarrow Y$ linear.

- (1) The graph of T is $\text{Gr}(T) := \{(x, Tx) \mid x \in X\}$.
- (2) The map T has a closed graph if $\text{Gr}(T)$ is closed in $X \times Y$.

Remark. Regardless if $X \times Y$ is given the product or box topology, the induced metrics are equivalent.

Proposition 1.6.1. *Let X, Y be normed spaces and $T : X \rightarrow Y$ be linear. The following are equivalent:*

- (1) T has a closed graph;
- (2) If $\{x_n\}_{n=1}^\infty$ is a sequence such that $x_n \rightarrow x$ and $Tx_n \rightarrow y$, then $y = Tx$.

Proof. Suppose T has a closed graph. Let $\{x_n\}_{n=1}^\infty$ be a sequence such that $x_n \rightarrow x$ and $Tx_n \rightarrow y$. Then $\{(x_n, Tx_n)\}_{n=1}^\infty \subseteq \text{Gr}(T)$ is a sequence such that $(x_n, Tx_n) \rightarrow (x, y)$. But since T has a closed graph, $(x, y) \in \text{Gr}(T)$. Hence $y = Tx$.

Conversely, if $\{(x_n, Tx_n)\}_{n=1}^\infty \subseteq \text{Gr}(T)$ is a sequence such that $(x_n, Tx_n) \rightarrow (x, y) \in X \times Y$, then $x_n \rightarrow x$ and $Tx_n \rightarrow y$. By hypothesis $y = Tx$, hence $(x, y) \in \text{Gr}(T)$. Thus T has a closed graph. $\square$

Remark. A linear map T having a closed graph is weaker than T being continuous.

Example 1.6.1. Recall the following standard result from an introductory analysis course: let $J \subseteq \mathbb{R}$ be a bounded interval and let $\{f_n\}_{n=1}^\infty \subseteq J^{\mathbb{R}}$ be a sequence of functions. Suppose

that there exists $x_0 \in J$ such that $\{f_n(x_0)\}_{n=1}^\infty$ converges, and that the sequence $\{f'_n\}_{n=1}^\infty$ of derivatives exists on J and converges uniformly on J to a function g . Then the sequence $\{f_n\}_{n=1}^\infty$ converges uniformly on J to a function f that has a derivative at every point of J and $f' = g$.

Indeed, define $D : C^1([0, 1]) \rightarrow C([0, 1])$ by $Df = f'$. Let $\{f_n\}_{n=1}^\infty \subseteq C^1([0, 1])$ be a sequence of functions such that $f_n \rightarrow f$ and $Df_n \rightarrow g$. Note that the convergence of $\{Df_n\}_{n=1}^\infty$ will be uniform by the compactness of $[0, 1]$. Hence by the above result, we have that $g = Df$. By Proposition 1.6.1, D has a closed graph. However, we showed in Example 1.3.3 that D is not continuous.

Definition 1.6.3. Let X, Y be normed linear spaces. A map $T : X \rightarrow Y$ is open if $T(U)$ is open in Y provided U is open in X .

Theorem 1.6.3 (Open-Mapping Theorem). *Let $T : X \rightarrow Y$ be a continuous, surjective, and linear mapping between Banach spaces. Then T is open.*

Remark. Many times throughout the following proof we will be using the fact that $B(0, n) = nB(0, 1)$.

Proof. Note that $Y = \bigcup_{n=1}^\infty T(nB_X(0, 1))$, hence $Y = \bigcup_{n=1}^\infty \overline{T(nB_X(0, 1))}$. Since Y is Banach, the Baire Category Theorem says there exists some $n \in \mathbb{N}$ such that $\overline{nT(B_X(0, 1))}$ has nonempty interior. So there exists $y_0 \in Y$ and $r > 0$ such that:

$$\begin{aligned} B_Y(y_0, r) &\subset \overline{nT(B_X(0, 1))} \\ &= \overline{nT(B_X(0, 1))}^3. \end{aligned}$$

Let $\epsilon > 0$. Since $y_0 \in B_Y(y_0, r)$, surely $y_0 \in \overline{nT(B_X(0, 1))}$. Hence there exists some $x_1 \in B_X(0, 1)$ such that $\|y_0 - nTx_1\|_Y < \epsilon$. Now let $y \in B(0, 1)$ be arbitrary. Then $y_0 + ry \in B_Y(y_0, r) \subset \overline{nT(B_X(0, 1))}$. Hence there exists $x_2 \in B_X(0, 1)$ such that $\|y_0 + ry - nTx_2\|_Y < \epsilon$. We can write:

$$\begin{aligned} y &= \frac{1}{r}(ry + y_0) - \frac{1}{r}y_0 \\ &= \frac{1}{r}(ry + y_0 - nTx_2) + \frac{n}{r}Tx_2 - \frac{1}{r}(y_0 - nTx_1) - \frac{n}{r}Tx_1 \\ &= \frac{1}{r}((y_0 + ry) - nTx_2) - \frac{1}{r}(y_0 - nTx_1) + \frac{n}{r}T(x_2 - x_1). \end{aligned}$$

We can rearrange the above equation as:

$$y - \frac{n}{r}T(x_2 - x_1) = \frac{1}{r}((y_0 + ry) - nTx_2) - \frac{1}{r}(y_0 - nTx_1).$$

³Follows from the fact that $\overline{B(x, r)} = \overline{B}(x, r)$ and the above remark.

Hence:

$$\begin{aligned} \left\| y - \frac{n}{r} T(x_2 - x_1) \right\| &= \left\| \frac{1}{r} ((y_0 + ry) - nTx_2) - \frac{1}{r} (y_0 - nTx_1) \right\| \\ &\leq \frac{1}{r} \|((y_0 + ry) - nTx_2)\| + \frac{1}{r} \|(y_0 - nTx_1)\| \\ &< \frac{2\epsilon}{r}. \end{aligned}$$

Take $x' := x_2 - x_1 \in B(0, 2)$. Then since $\left\| y - \frac{n}{r} T(x') \right\| < \frac{2\epsilon}{r}$ can be made arbitrarily small, we have that $y \in \overline{T(B_X(0, \frac{2n}{r}))}$. Take $\lambda := \frac{2n}{r}$. Thus we've established the inclusion $B_Y(0, 1) \subset \overline{TB_X(0, \lambda)}$.

Let $y \in B_Y(0, 1) \subset \overline{T(B_X(0, \lambda))}$. Find $x_1 \in B_X(0, \lambda)$ such that $\|y - Tx_1\|_Y < \frac{1}{2}$. Then:

$$\begin{aligned} y - Tx_1 &\in B_Y\left(0, \frac{1}{2}\right) \\ &= \frac{1}{2} B_Y(0, 1) \\ &\subset \frac{1}{2} \overline{T(B_X(0, \lambda))} \\ &= \overline{\frac{1}{2} T(B_X(0, \lambda))} \\ &= \overline{T\left(B_X\left(0, \frac{\lambda}{2}\right)\right)}. \end{aligned}$$

Now pick $x_2 \in B_X(0, \frac{\lambda}{2})$ such that $\|y - Tx_1 - Tx_2\| < \frac{1}{2^2}$. Then:

$$\begin{aligned} y - Tx_1 - Tx_2 &\in B_Y\left(0, \frac{1}{2^2}\right) \\ &= \frac{1}{2^2} B_Y(0, 1) \\ &\subset \frac{1}{2^2} \overline{T(B_X(0, \lambda))} \\ &= \overline{\frac{1}{2^2} T(B_X(0, 1))} \\ &= \overline{T\left(B_X\left(0, \frac{\lambda}{2^2}\right)\right)}. \end{aligned}$$

Inductively, pick $x_k \in B_X(0, \frac{\lambda}{2^{k-1}})$ such that $\left\| y - \sum_{j=1}^k Tx_j \right\| < \frac{1}{2^k}$. Note that for all $k \in \mathbb{N}$, we have $\|x_k\|_X < \frac{\lambda}{2^{k-1}}$. Therefore $\sum_{j=1}^{\infty} x_j$ is absolutely convergent. Since X is Banach, by

Proposition 1.1.2 the series $\sum_{j=1}^{\infty} x_j$ is convergent. In particular:

$$\begin{aligned} \left\| \sum_{j=1}^{\infty} x_j \right\|_X &\leq \sum_{j=1}^{\infty} \|x_j\|_X \\ &< \sum_{j=1}^{\infty} \frac{\lambda}{2^{j-1}} \\ &= 2\lambda. \end{aligned}$$

Hence $\sum_{j=1}^{\infty} x_j \in B_X(0, 2\lambda)$. Now note that $\sum_{j=1}^k x_j \rightarrow \sum_{j=1}^{\infty} x_j$ in X . Also, $\left\| y - \sum_{j=1}^k T x_j \right\| < \frac{1}{2^k}$ implies that $T \left(\sum_{j=1}^k x_j \right) \rightarrow y$ in Y . Since T is continuous, T has a closed graph. Hence $y = T \sum_{j=1}^{\infty} x_j \in T(B_X(0, 2\lambda))$. Since $y \in B_Y(0, 1)$ was arbitrary, we've established the inclusion $B_Y(0, 1) \subset T(B_X(0, 2\lambda))$.

Now let $U \subseteq X$ be open. Let $y_0 \in T(U)$. Since T is surjective, find $x_0 \in U$ such that $T x_0 = y_0$. Since U is open there exists an $\epsilon > 0$ such that $B_X(x_0, \epsilon) \subseteq U$. Therefore:

$$\begin{aligned} T(U) &\supset T(B_X(x_0, \epsilon)) \\ &= T x_0 + \frac{\epsilon}{2\lambda} T(B_X(0, 2\lambda)) \\ &\supset y_0 + \frac{\epsilon}{2\lambda} B_Y(0, 1) \\ &= B_Y\left(y_0, \frac{\epsilon}{2\lambda}\right) \end{aligned}$$

Thus $T(U)$ is open, establishing T as an open map. $\square$

Corollary 1.6.3. *Let $T : X \rightarrow Y$ be a bijective, linear, and continuous map between Banach spaces. Then T^{-1} is continuous.*

Proof. Since T is continuous, T has a closed graph. By the Open-Mapping Theorem T is an open map. Hence $(T^{-1})^{-1}(U) = T(U)$ is open. $\square$

Corollary 1.6.4. *Let X be a vector space. Suppose $\|\cdot\|_{\alpha}$ and $\|\cdot\|_{\beta}$ are norms such that $(X, \|\cdot\|_{\alpha})$ and $(X, \|\cdot\|_{\beta})$ are Banach. If there exists a $C > 0$ such that $\|x\|_{\alpha} \leq C \|x\|_{\beta}$ for all $x \in X$, then there exists $C' > 0$ such that $\|x\|_{\beta} \leq C' \|x\|_{\alpha}$ for all $x \in X$.*

Proof. Certainly the map $\text{id} : (X, \|\cdot\|_{\beta}) \rightarrow (X, \|\cdot\|_{\alpha})$ is bijective and linear. Moreover, $\|\text{id } x\|_{\alpha} = \|x\|_{\alpha} \leq C \|x\|_{\beta}$. Hence id is continuous. By Corollary 1.6.3 id^{-1} is continuous. Hence there exists $C' > 0$ such that $\|x\|_{\beta} = \|\text{id}^{-1} x\|_{\beta} \leq C' \|x\|_{\alpha}$. $\square$

Remark. The above corollary can be rephrased as: if $\|\cdot\|_{\alpha}$ and $\|\cdot\|_{\beta}$ are norms such that $(X, \|\cdot\|_{\alpha})$ and $(X, \|\cdot\|_{\beta})$ are Banach, then $\|\cdot\|_{\alpha}$ and $\|\cdot\|_{\beta}$ are equivalent.

Theorem 1.6.4 (Closed Graph Theorem). *Let X and Y be Banach spaces. If $T : X \rightarrow Y$ is linear and has a closed graph, then T is continuous.*

Proof. Define $\|\cdot\|_* : X \rightarrow \mathbb{R}$ by $\|x\|_* := \|x\|_X + \|Tx\|_Y$. It is routine to check that $(X, \|\cdot\|_*)$ is a normed linear space. We will check that $(X, \|\cdot\|_*)$ is Banach. Suppose $\{x_k\}_{k=1}^\infty$ is Cauchy with respect to $\|\cdot\|_*$. Then clearly $\{x_k\}_{k=1}^\infty$ is Cauchy with respect to $\|\cdot\|_X$ and $\{Tx_k\}_{k=1}^\infty$ is Cauchy with respect to $\|\cdot\|_Y$. Since X and Y are Banach, there exists $x \in X$ and $y \in Y$ such that $x_k \rightarrow x$ with respect to $\|\cdot\|_X$ and $Tx_k \rightarrow y$ with respect to $\|\cdot\|_Y$. Since T has a closed graph, $y = Tx$. Hence:

$$\begin{aligned} \|x - x_k\|_* &= \|x - x_k\|_X + \|Tx - Tx_k\|_Y \\ &\rightarrow 0 \text{ as } k \rightarrow \infty. \end{aligned}$$

Thus $x_k \rightarrow x$ with respect to $\|\cdot\|_*$, establishing that $(X, \|\cdot\|_*)$ is Banach. But since $(X, \|\cdot\|_X)$ and $(X, \|\cdot\|_*)$ are Banach, and since $\|x\|_X \leq \|x\|_*$, Corollary 1.6.4 says there exists $C' > 0$ such that $\|x\|_* \leq C' \|x\|_X$. Therefore $\|Tx\|_Y \leq \|x\|_* \leq C' \|x\|_X$ for all $x \in X$. Thus T is continuous. $\square$

Example 1.6.2.

Example 1.6.3.

Exercises

Exercise 1.6.1. *Let f be infinitely differentiable on $\mathbb{R}$. Suppose for any $x \in \mathbb{R}$ there is some n such that $f^{(n)}(x) = 0$ (note that n may depend on x). Show that f must be a polynomial. Hint: Consider the sets $Z_n = \{x \mid f^{(n)}(x) = 0\}$ and recall Baire's Category Theorem.*

Proof.

$\square$

Exercise 1.6.2. *Let $T : X \rightarrow Y$ be a linear map between normed spaces. Prove the assertion made in class: The graph $\{(x, Tx) \mid x \in X\}$ of T is closed in $X \times Y$ means:*

$$\text{if } x_n \rightarrow x \text{ and } Tx_n \rightarrow y \text{ then } y = Tx.$$

Proof.

$\square$

Exercise 1.6.3. *Define $T : c_0 \rightarrow c_0$ by $T(\{x_n\}_{n=1}^\infty) = \{nx_n\}_{n=1}^\infty$. Determine if T is injective, surjective, open, closed, and/or invertible.*

Proof.

□

Exercise 1.6.4. Let $L : X \rightarrow Y$ be a linear map between Banach spaces. Suppose that the conditions $x_n \rightarrow 0$ and $Lx_n \rightarrow y$ imply that $y = 0$. Prove that L is continuous.

Proof. Let $\{(x_n, Tx_n)\}_{n=1}^{\infty} \subseteq \text{Gr}(L)$ be a sequence such that $(x_n, Tx_n) \rightarrow (x, y) \in X \times Y$. Then $x_n \rightarrow x$ and $Tx_n \rightarrow y$. Define $u_n := x_n - x$. Then $u_n \rightarrow 0$ and $Tu_n \rightarrow y - Tx$. By hypothesis $y - Tx = 0$, giving $y = Tx$. Thus $(x, y) \in \text{Gr}(T)$, which means T has a closed graph. By the Closed Graph Theorem, T is continuous. □

Exercise 1.6.5. Prove that if a closed map has an inverse then the inverse is also closed.

Proof. Let $\{y_n\}_{n=1}^{\infty}$ be a sequence in Y such that $y_n \rightarrow y$ and $T^{-1}y_n \rightarrow x$. Since T admits an inverse, it is bijective. Hence for each $y_n \in Y$, there exists $x_n \in X$ such that $y_n = Tx_n$. The two sequences above can be rewritten as $Tx_n \rightarrow y$ and $T^{-1}(Tx_n) = x_n \rightarrow x$. Since T has a closed graph, $y = Tx$. Since T is bijective, $x = T^{-1}y$. Thus T^{-1} has a closed graph. □

Exercise 1.6.6. Let $L : X \rightarrow Y$ be a continuous linear surjection, where X and Y are Banach spaces. Let $y_n \rightarrow y$ in Y . Prove that there exists points $x_n \in X$ and a constant $c \in \mathbb{R}$ such that $Lx_n = y_n$, the sequence $\{x_n\}_{n=1}^{\infty}$ converges, and $\|x_n\| \leq c \|y_n\|$.

Proof.

□